

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- a) Explain Harvard architecture and give reason behind this architecture being faster than Von-Neumann architecture. 7
- b) Define flags and addressing modes. What are the different addressing modes used in 8085 microprocessor? Explain with examples. 8
- a) Draw the timing diagrams for the instruction IN 4FH. 8
- b) Write a 8085 program to find the largest of three numbers. 7
- a) Write an 8085 ALP to find ODD and EVEN number among six bytes of data: 23H, 41H, 56H, AFH, C5H & A7H stored in the memory location starting from 2500H and place the ODD and EVEN numbers in memory location starting from 2600H and 2400H respectively. 8
- b) Design a circuit to interface a 8 KB ROM to 8085 microprocessor. The starting address being 1000H. 7
- a) Draw the functional block diagram of 8086 Microprocessor and explain how segmentation and pipelining concepts are implemented in this microprocessor. 8
- b) Write an assembly language programming for 8086 to find the square root of a given number. Assume that a number is of two digits and is perfect square. 7
- a) Specify the functions of the following 8086 instructions: AAA, LDS, DIV, MOVSB, NOT, CALL, ROL. 7
- b) Define IVT. Explain interrupt vector table for 8086 microprocessor. 8
- a) Explain the RS232C standard for serial communication. Explain its advantages and disadvantages. 7
- b) Explain the functional block diagram of 8259 PIC. 8
7. Write short notes on: (Any two) 2x5
 - a) Synchronous and Asynchronous data bus
 - b) Modes of operation of 8255
 - c) Memory mapped I/O and I/O mapped I/O...